

Machine Learning for Exoplanet Detection: Identifying Exoplanets in Light Curves

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General background of Exoplanet discovery

Kepler/K2 mission

- 2009-2018
- Continuously monitored one section of sky for 4 years
- More than 4000 potential planets, and 2700 are confirmed

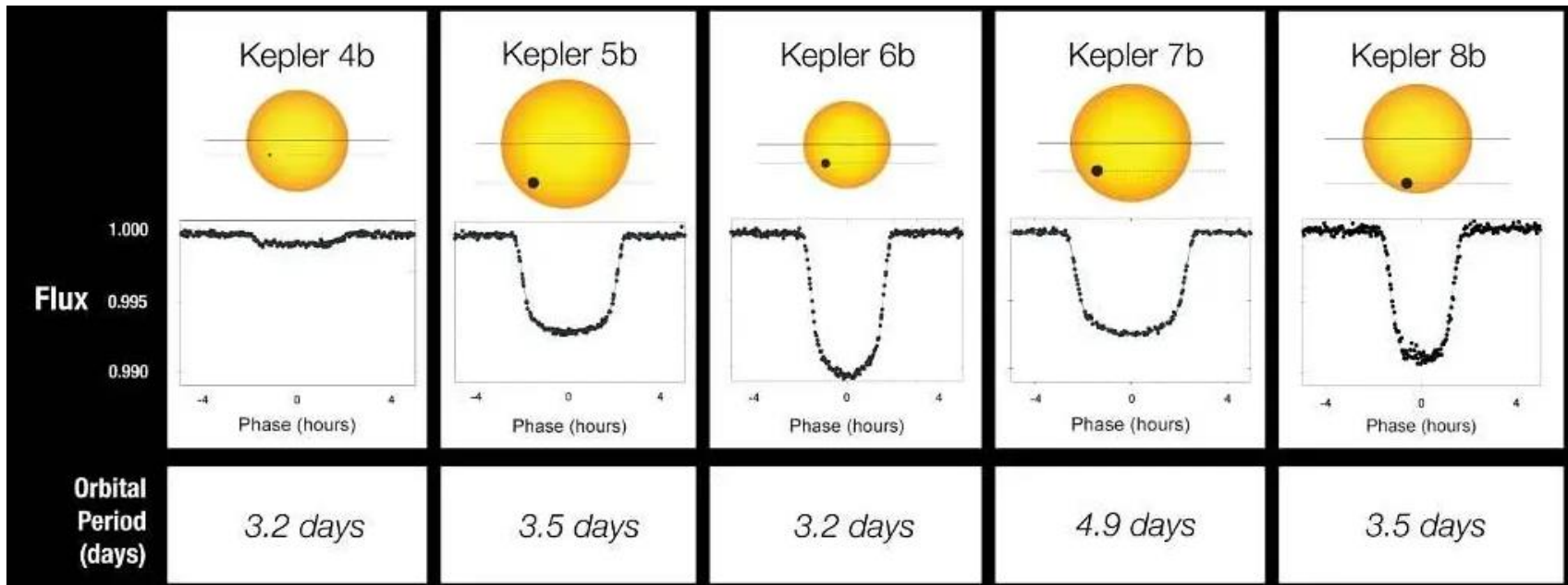
TESS (Transiting Exoplanet Survey Satellite) mission

- 2018-present
- Monitors a different segment of sky every 27 days
- Over 7000 potential planets, and almost 700 confirmed

Exoplanets often found by analyzing light curve data

- Looking for periodic threshold crossing events, or dips in intensity of the light coming from the star

What are Threshold Crossing Events?



NASA. (2010, January 4). Light curves of Kepler's first 5 discoveries [Image]. NASA.
<https://www.nasa.gov/image-article/light-curves-of-keplers-first-5-discoveries/>

Problem Statement and Method

There is currently way more data than can be analyzed by hand, and Machine Learning (ML) is the logical solution.

We will compare the performance of the following ML methods after training and testing on the same sets of data:

- Convolutional Neural Network (CNN)
- Transformers
- Recursive Neural Network (RNN)

We will use these deep learning methods to analyze light curve data for threshold crossing events (TCEs) which may indicate the transit of an exoplanet

Data sets

- **Simulated K2 data** (free data sets available from NASA Exoplanet Archive)
- **Labeled TESS light curves** (list of confirmed planets and false positives from NASA Exoplanet Archive, light curves available from MAST)
- Reserve 20% of each data set for testing
- If models are accurate, compare results when looking at unlabeled Kepler/K2/TESS data

CNN Research Papers

Shallue, C. J., & Vanderburg, A. (2018). Identifying exoplanets with deep learning: A five-planet resonant chain around Kepler-80 and an eighth planet around Kepler-90. *The Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>

Dattilo, A., Vanderburg, A., Shallue, C. J., Mayo, A. W., Berlind, P., Bieryla, A., Calkins, M. L., Esquerdo, G. A., Everett, M. E., Howell, S. B., Latham, D. W., Scott, N. J., & Yu, L. (2019). Identifying exoplanets with deep learning II: Two new super-Earths uncovered by a neural network in K2 data. *The Astronomical Journal*, 157(5), 169. <https://doi.org/10.3847/1538-3881/ab0e12>

Osborn, H. P., Ansdell, M., Ioannou, Y., Sasdelli, M., Angerhausen, D., Caldwell, D., Jenkins, J. M., Räissi, C., & Smith, J. C. (2020). Rapid classification of TESS planet candidates with convolutional neural networks. *Astronomy & Astrophysics*, 633, A53. <https://doi.org/10.1051/0004-6361/201935345>

Identifying Exoplanets with Deep Learning (The Astronomical Journal, 2018)

Goal: Find and build a classifier for Kepler data

Approach: Compared linear, fully connected, and 1-D CNN models using Kepler and DR24 data. Best results from CNN using global and local views of light curve sequences.

Techniques: Phase-folding, spline detrending, augmentation, dropout, ensemble learning.

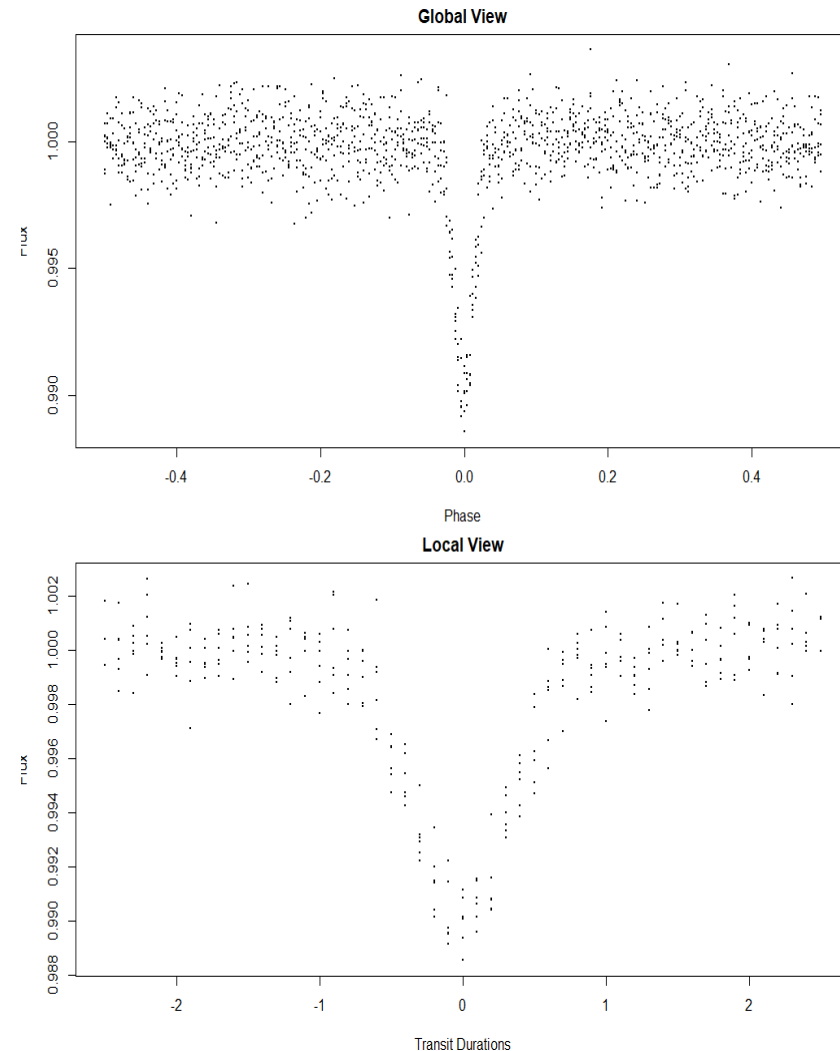
Model (AstroNet): 1-D CNN with 4/10 Convolutional layers, Max Pooling, Dense FC-512 layer with a sigmoid probability output.

Results Observational: 96% Accuracy, 98.8% AUC, 96% precision, 95% recall.

Results Simulated: 81% recovery of true planets, misclassified 90% background planets, and misclassified 65% of eclipsing binaries.

Findings: Kepler-80g and Kepler-90i

Takeaways: Models limited by generalization. Extend to simulated data and centroid features.



Identifying Exoplanets with Deep Learning II (The Astronomical Journal, 2019)

Goal: Build a CNN classifier for K2 data

Approach: Extended AstroNet architecture to handle K2 dataset. Trained on K2 TCEs and ran validation and tuning on simulated TCEs. Used local and global views.

Techniques: Phase-folding, spline detrending, data augmentation, ensemble learning, scalar features (R_p/R^* and impact).

Model (AstroNet-K2): 1-D CNN with 4/8 Convolutional layers, Max Pooling, Dense FC-512 layer with a sigmoid probability output.

Results Observational: Accuracy 97.8%, AUC 98.8%, 94% in capturing planets.

Findings: K2-293B and K2-294B

Takeaways: Shows CNNs can work on noisier light curve sequences and that scalar features may help in improving classification.



Rapid Classification of TESS Planet Candidates with Convolutional Neural Networks (Astronomy & Astrophysics, 2019)

Goal: Build a CNN classifier for TESS data

Approach: Adapted AstroNet/ExoNet CNN. Trained on TSOP-301 (simulated). Used local and global views with centroids.

Techniques: Phase-folding, normalization, balanced batches, augmentation, cross-validation

Model: 1-D CNN with 4/8 Convolutional layers, Max Pooling, Dense FC-256 layer into both a sigmoid (binary) and softmax(multi-class) output

Results Simulated: AP(Acc/Prec) of 97.3%

Results Observational: Planet Precision of 91.8% and 61% recovery on real TESS exoplanets

Takeaways: Proof that CNNs can reduce human vetting and classify planets though possibly limited in generalization.

Transformer model

- Input/output embedding
 - Creates tokens and encodes positional data
 - Encoder/decoder
 - Creates relationships between tokens
 - Multi-Head Attention
 - Weights the relationships between tokens
 - Each token is compared to every other token
 - Each head compares the tokens in a different way
 - Relationship weights are determined by the query, key, and value vectors
 - Add & Norm
 - Keeps important information distinct and appropriately scaled
 - Feed Forward
 - Creates new token vectors by applying learned weights for the next step
-
- Good for sequential data (text or time series)
 - Initially used in Large Language Models
 - Parallelization makes training more efficient

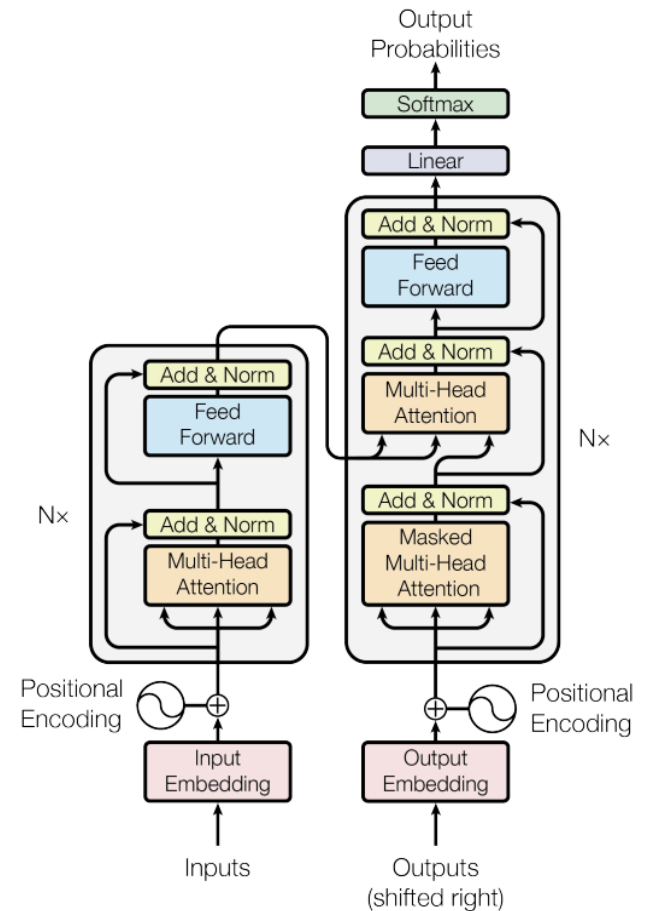


Figure 1: The Transformer - model architecture.

Figure 1 reproduced with permission by Google for scholarly use, from Vaswani et al. (2017).

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS 2017)* (pp. 6000–6010). Curran Associates Inc. <https://arxiv.org/pdf/1706.03762>

Transformer Research Papers

- Morvan, M., Nikolaou, N., Yip, K. H., & Waldmann, I. (2022). Don't pay attention to the noise: Learning self-supervised representations of light curves with a denoising time series transformer. *Proceedings of the Thirty-ninth International Conference on Machine Learning (ICML 2022)*. <https://arxiv.org/abs/2206.08447>
- Salinas, H., Pichara, K., Brahm, R., Pérez-Galarce, F., & Mery, D. (2023). Distinguishing a planetary transit from false positives: A Transformer-based classification for planetary transit signals. *Monthly Notices of the Royal Astronomical Society*, 522(3), 3201–3216. <https://doi.org/10.1093/mnras/stad1173>
- Salinas, H., Brahm, R., Olmschenk, G., Barry, R. K., Pichara, K., Silva, S. I., & Araujo, V. (2025). Exoplanet transit candidate identification in TESS full-frame images via a transformer-based algorithm. *Monthly Notices of the Royal Astronomical Society*, 538(3), 2031–2049. <https://doi.org/10.1093/mnras/staf347>

Don't pay attention to the noise: Learning self-supervised representations of light curves with a denoising time series transformer

Goal: To create a self-supervised framework that remove noises and repopulates data gaps (imputation) from raw light curves

Data set: TESS (raw light curve data)

Method: Denoising Time Series Transformer (DTST)

Used 30% masking ratio on light curve data

Encoder-only, with three layers and eight heads each

Results: Lowest residual noise (IQR 0.385%) and lowest error (MAE 0.235%) compared to two other methods

Impact: Does not need a fixed sliding window size, greater adaptability to data

Morvan, M., Nikolaou, N., Yip, K. H., & Waldmann, I. (2022). Don't pay attention to the noise: Learning self-supervised representations of light curves with a denoising time series transformer. *Proceedings of the Thirty-ninth International Conference on Machine Learning (ICML 2022)*. <https://arxiv.org/abs/2206.08447>

Distinguishing a planetary transit from false positives: A Transformer-based classification for planetary transit signals

Goal: Classify Threshold Crossing Events (TCEs) as either true exoplanet transits or false positives

Data set: TESS phase-folded light curves, centroid data

Method: CNN tokenization, then positional data embedding, then three encoder layers, each containing four multi-head self-attention layers, with four heads each

Results: Achieved competitive performance with CNNs with average precision, recall, and $F1 \approx 0.88$

Impact: Transformers can match CNN accuracy

Salinas, H., Pichara, K., Brahm, R., Pérez-Galarce, F., & Mery, D. (2023). Distinguishing a planetary transit from false positives: A Transformer-based classification for planetary transit signals. *Monthly Notices of the Royal Astronomical Society*, 522(3), 3201–3216.
<https://doi.org/10.1093/mnras/stad1173>

Exoplanet transit candidate identification in *TESS* full-frame images via a transformer-based algorithm

Goal: Identify exoplanet candidates in TESS Full Frame Images (FFIs) without relying on phase folding or periodicity assumptions

Data set: TESS raw full-frame images, centroid data, background flux time series, artificially injected light curves

Method: 2-layer CNN data tokenization, then positional embedding, then one encoder with four multi-head layers, each with four heads

Results: Average AUC of 0.88 and F1 score of 0.82. Then observed more than 4 million light curves, indicating 0.1% were possible candidates, 25% of which were valid candidates

Impact: Successfully identified 214 new planetary candidates, including 88 single-transit candidates, indicates high ability to generalize

Salinas, H., Brahm, R., Olmschenk, G., Barry, R. K., Pichara, K., Silva, S. I., & Araujo, V. (2025). Exoplanet transit candidate identification in TESS full-frame images via a transformer-based algorithm. *Monthly Notices of the Royal Astronomical Society*, 538(3), 2031–2049.
<https://doi.org/10.1093/mnras/staf347>

RNN Research Papers

Vida, K., Bódi, A., Szklenár, T., & Seli, B. (2021). Finding flares in Kepler and TESS data with recurrent deep neural networks. *Astronomy & Astrophysics*, 652, A107.

Kügler, S. D., Gianniotis, N., & Polsterer, K. L. (2016). An explorative approach for inspecting Kepler data. *Monthly Notices of the Royal Astronomical Society*, 455(4), 4399–4405.

Du, N., Dai, H., Trivedi, R., Upadhyay, U., Gomez-Rodriguez, M., & Song, L. (2016). Recurrent marked temporal point processes: Embedding event history to vector. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1555–1564). New York, NY: ACM.

Finding flares in Kepler and TESS data with recurrent deep neural networks

Why RNN: Long, noisy light curves → need memory of past context.

Data / tools: Kepler (train) → TESS (test); automated flare detection..

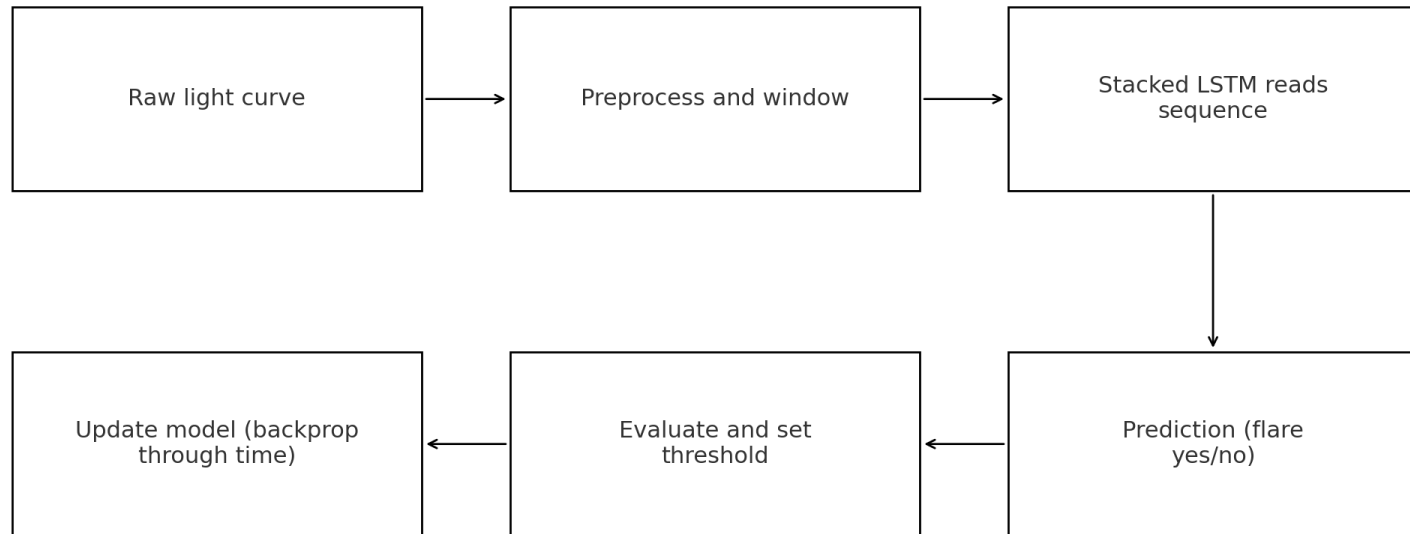
RNN used: Stacked **LSTM** + dropout; sigmoid output (binary).

Why this RNN: LSTM > GRU/CNN here for temporal dependencies.

Success: ~80–90% precision/recall for strong flares; **Kepler**→**TESS** transfer.

My takeaway: Start with 2–3×LSTM (128–256), class weighting, dropout, early stopping; include hard negatives.

LSTM (Vida et al., 2021) — simple flow



An explorative approach for inspecting Kepler data

Why RNN/seq model: Capture full **sequence dynamics** of Kepler curves.

Data / tools: Kepler; **ESN** reservoir + **autoencoder** (sequence-aware).
Success: Better clustering/visualization of dynamics than PCA/plain AE.

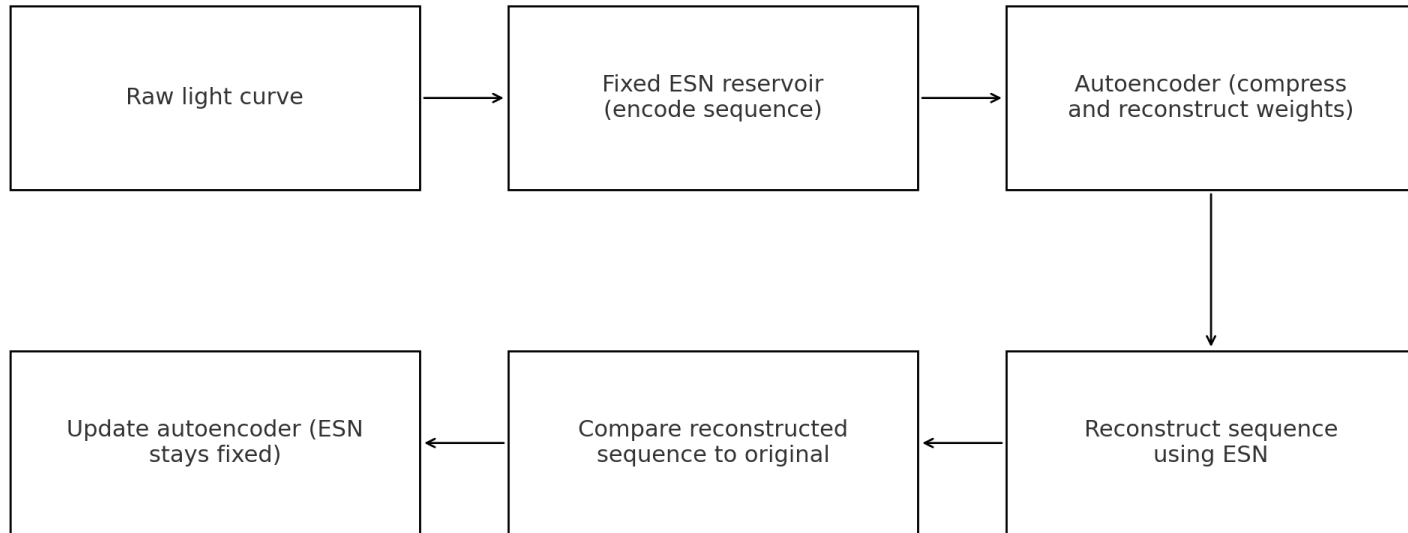
RNN used: **Echo State Network (ESN)** reservoir + **autoencoder** (ESN-AE).

Why this RNN: ESN gives fast recurrent state; optimize in **sequence space**.

Success: Clearer clustering/visualization vs. PCA/plain AE (qualitative SOTA).

My takeaway: Use **sequence-aware diagnostics/objectives**; inspect scores over time, not just final embeddings.

ESN-Autoencoder (Kügler et al., 2016) — simple flow



Recurrent marked temporal point processes: Embedding event history to vector

Why RNN: Need model of **event timing** + type, not just labels.

Data / tools: General event streams; **RMTPP** with RNN history encoder.

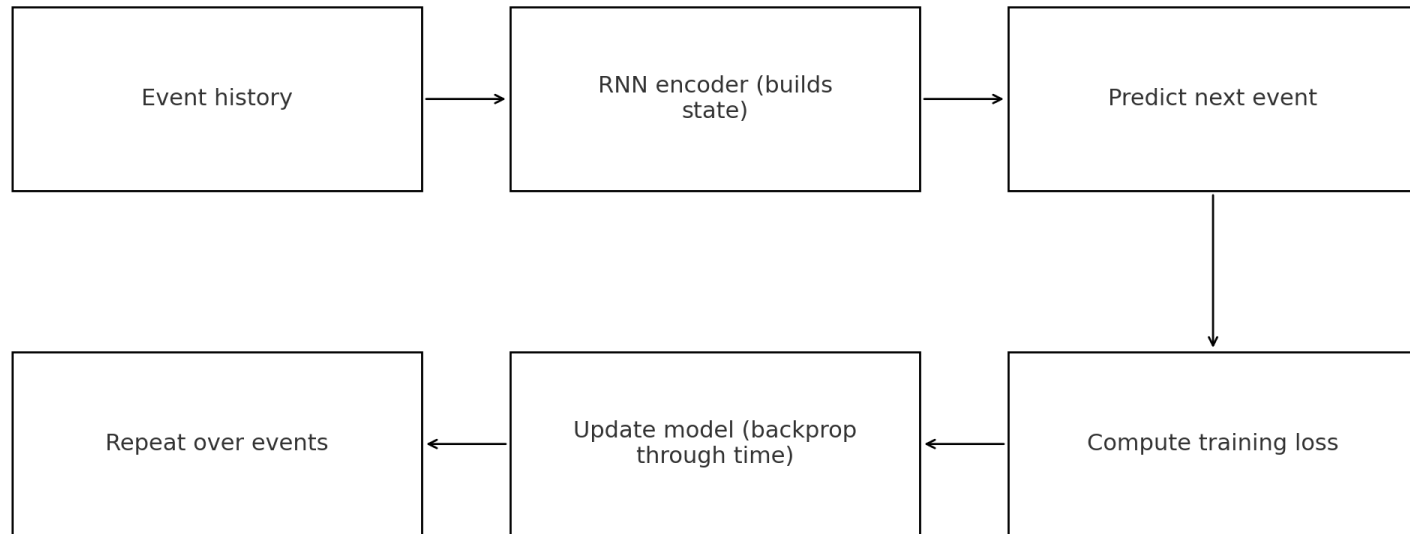
RNN used (specific): **RMTPP** with an **RNN history encoder** (tanh-RNN in the formulation) to parametrize intensity.

Why this RNN: Non-parametric timing from history; flexible vs. fixed forms.

Success: Better next-event time/type prediction across domains.

My takeaway: Add **timing helper** (intervals / rough period) as feature or tiny aux loss to favor periodic transits.

RMTTP (Du et al., 2016) — simple flow



Considerations in achieving goals

- Determining sky sector?
- Preprocessing data
 - Standardization and normalization
- Creating test data
- Ruling out false positives
- Finding uniform analysis metric to compare methods

Project Plan

- Goal: Identify Kepler/TESS TCEs
- Data: Kepler/TESS/Simulated light curves
- Metrics: Accuracy, Precision, Recall, F1, and AUC (Area Under Curve)
- Milestones:
 - Week 1-2: Preprocess light curves
 - Week 3-4: Train model
 - Week 5-6: Evaluate using precision, recall, AUC
 - Week 7-8: Compare models and present findings



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